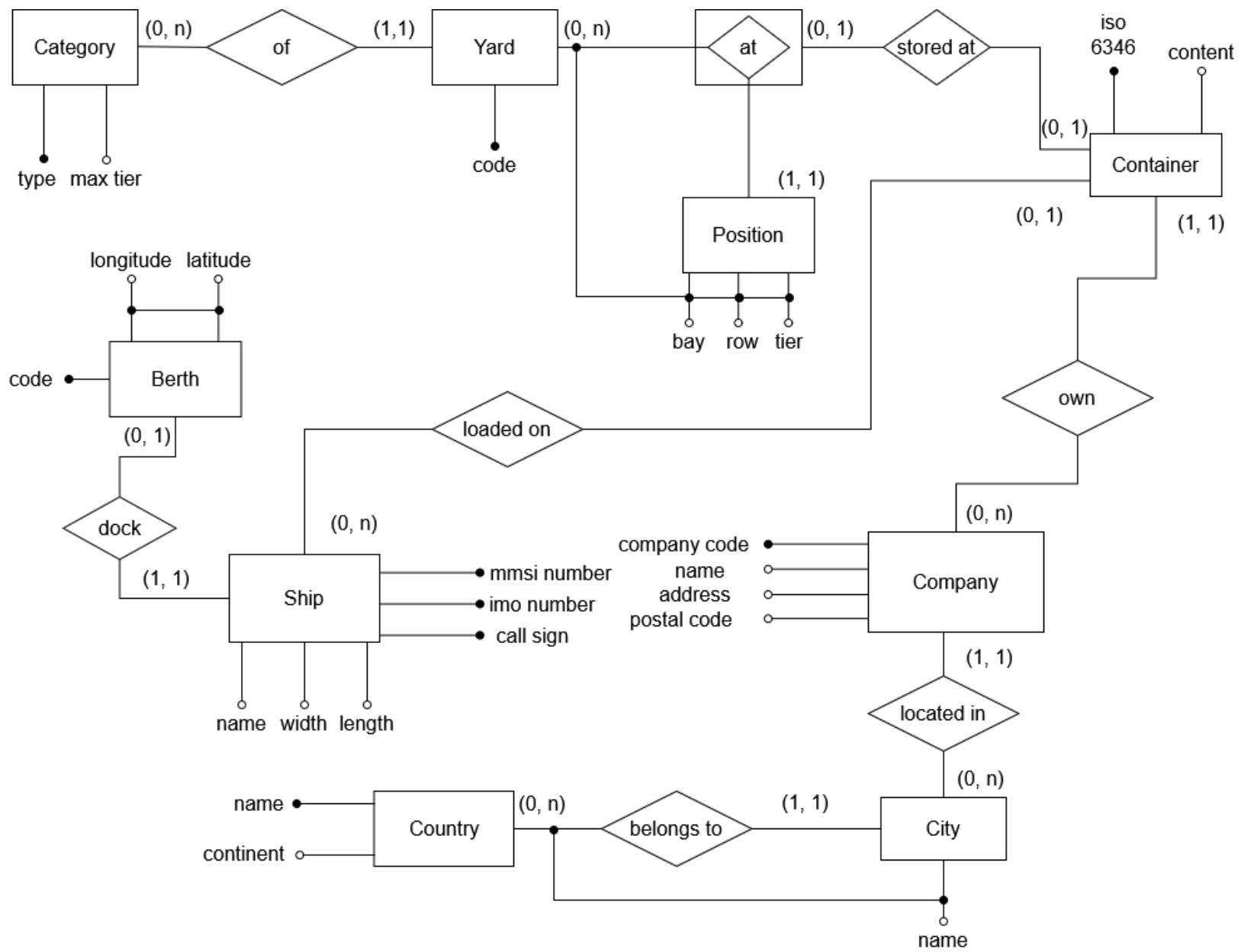




Group Number: T37

Entities + Attributes:

Container:

- Container reflects the specification that the application stores and manages information about containers.
- As reflected in the specification, the identity will be the ISO 6346 identification and we assume this is globally unique.

Company

- Company reflects the specification that the application stores and manages information about companies that own containers.
- As each ship is identified by a unique MMSI number of which a substring is the company code, we also assume that the company code, e.g. "MSC", is also unique and therefore the identity of the company entity.
- We assume that multiple companies can share the same address; thus, address is not an identity.

City

- City reflects the specification that the database is to store and manage information about cities.
- City name consists of its name and the national region in which it is in (e.g. Springfield, Oregon). We assume that the city entity is uniquely identified by its name within each country (further elaborated below); therefore, the city name is a partial identity.
- In the event that a country has no city, the city and region name is the country name (e.g. Singapore, Singapore)

Country

- Country reflects the specification that the database is to store and manage information about countries.
- We assume that the country name is globally unique as reflected by the current state of the world and thus is the identity of the country entity.

Ship

- Ship reflects the specification that the database is to store and manage information about ships.
- As stated in the specification that the MMSI, IMO, and call sign of a ship are each globally unique; they are each identities of the Ship entity.

Berth

- Berth reflects the specification that the database is to store and manage information about berths.

- As stated in the specification that each berth is identified by code, code is an identity of the entity. As the berth is also identified by position as stated in the specification, the longitude and longitude of the berth is a composite identity.

Yard

- Yard reflects the specification track yards in the port.
- As stated in the specification that each yard is identified by a unique code, the yard code is the identity.

Category

- Category reflects the specification that types of yards should be recorded.
- We will assume that the yard type is uniquely identifiable by name (e.g. as specified, the AI-enabled yard), and thus is the identity.
- Given that each yard has a maximum tier, we also record this as an attribute.

at

- As each Container may be stored in a specific position in a specific yard, we use the `at` relation between Yard and Position as an aggregate entity that relates to Container through `stored at`.

Relations:

of: Category (0,n) — Yard (1,1)

- Every yard has exactly one category. A category may currently have no yards, e.g. a category that has been designed but not built yet.

at: Yard (0,n) — Position (1,1)

- Every position belongs to exactly one yard. A yard that is not currently fully built may not have any positions to put the container in. Position is identified within its yard by (bay, row, tier), which is represented as aggregate.

stored at: at (aggregate) (0,1) — Container (0,1)

- In this case we have an aggregate with the relation `at`, which represents a position within a specific yard. The min constraint for `at` is 0, representing a position in a yard that has no containers at the moment. At the same time, a position in a yard can only hold at most one container.
- A position may be empty and can hold at most one container. A container participates optionally because it may instead be loaded on a ship. Similarly, a container can be stored at at most one position at a time.

loaded on: Ship (0,n) — Container (0,1)

- A ship may carry zero to as many containers as its capacity. A container can be loaded on at most one ship, or stored in a yard.

dock: Berth (0,1) — Ship (1,1)

- A berth can be empty, but cannot hold more than one ship at a time. Every recorded ship must be docked at exactly one berth, and we don't record ships that are not docked at a berth.

own: Company (0,n) — Container (1,1)

- Every container has exactly one owner. A company can own zero to many containers.

located in: Company (1,1) — City (0,n)

- Every company has exactly 1 central headquarter (if there are multiple, we should choose just 1), i.e. belonging to exactly 1 city. A city can exist even if no company is located there, and a city can hold many companies.

belongs to: City (1,1) — Country (0,n)

- Every city belongs to exactly one country. A country may be stored even if it currently has no cities, and there may be multiple cities with the same name in a different country. Note that for a country that has multiple cities with the same name, we shall append the city's region to its name to distinguish them, (e.g. Springfield, Oregon).

Constraints that are not enforced by our ERD

1. The tier of a container stored at a yard must be between 1 and max tier.
2. For every container, there must be a container in the same bay in the same row for every tier below this container's tier.
3. A container is either positioned in a yard or on a ship.
4. The first 3 letters of the ISO code of a container should match with the first 3 letters of the company code.